Problem Set 3

1) $u(x) = (x_1^\rho + x_2^\rho)^{1/\rho}$

a) $\max\limits_{x_1, x_2} \; -P_1 x_1 - P_2 x_2 \quad s.t. \quad (x_1^\rho + x_2^\rho)^{1/\rho} = u_0$

$$\mathcal{L} = -P_1 x_1 - P_2 x_2 + \lambda\left((x_1^\rho + x_2^\rho)^{1/\rho} - u_0\right)$$

C.P.O.

$$[x_1]: \; -P_1 + \lambda \frac{1}{\cancel{\rho}} (x_1^\rho + x_2^\rho)^{\frac{1}{\rho}-1} \cancel{\rho}\, x_1^{\rho-1} = 0 \quad (1)$$

$$[x_2]: \; -P_2 + \lambda \frac{1}{\cancel{\rho}} (x_1^\rho + x_2^\rho)^{\frac{1}{\rho}-1} \cancel{\rho}\, x_2^{\rho-1} = 0 \quad (2)$$

$$[\lambda]: \; u_0 = (x_1^\rho + x_2^\rho)^{1/\rho} \quad (3)$$

$$\frac{(1)}{(2)}: \; \frac{x_1^{\rho-1}}{x_2^{\rho-1}} = \frac{P_1}{P_2} \Rightarrow x_2^{1-\rho} = \frac{P_1}{P_2} x_1^{1-\rho}$$

$$u_0^\rho = x_1^\rho + \left(\frac{P_1}{P_2} x_1^{1-\rho}\right)^{\rho/1-\rho}$$

$$u_0^\rho = x_1^\rho + \left(\frac{P_1}{P_2}\right)^{\rho/1-\rho} \cdot x_1^\rho = x_1^\rho \left(\frac{P_2^{\frac{\rho}{1-\rho}} + P_1^{\frac{\rho}{1-\rho}}}{P_2^{\rho/1-\rho}}\right) \qquad r = \frac{\rho}{1-\rho}$$

$$x_1^h = u_0 \left(\frac{P_2^r}{P_2^r + P_1^r}\right)^{1/\rho} ; \; \text{Por simetría} \Rightarrow x_2^h = u_0 \left(\frac{P_1^r}{P_1^r + P_2^r}\right)^{1/\rho}$$

b) La demanda Marshalliana viene dada por:

$$X_i^*(P,y) = \frac{P_i^{\frac{1}{\rho-1}} \cdot y}{P_1^{\frac{\rho}{\rho-1}} + P_2^{\frac{\rho}{\rho-1}}}$$

$$V(P,y) = \left(X_1^*(P,y)^\rho + X_2^*(P,y)^\rho\right)^{1/\rho}$$

$$= \left(\frac{P_1^{-r} y^\rho + P_2^{-r} y^\rho}{(P_1^{-r} + P_2^{-r})^\rho}\right)^{1/\rho}$$

$$= y \; \frac{(P_1^{-r} + P_2^{-r})^{1/\rho}}{P_1^{-r} + P_2^{-r}}$$

$$= y \; (P_1^{-r} + P_2^{-r})^{1/r}$$

c) $V(P, e(P, u_0)) = u_0$

$$V(P, e(P, u_0)) = e(P, u_0)\left(P_1^{-r} + P_2^{-r}\right)^{1/r} = u_0$$

$$e(P, u_0) = \frac{u_0}{(P_1^{-r} + P_2^{-r})^{1/r}}$$

$$= u_0 \, (P_1^{-r} + P_2^{-r})^{-1/r}$$

d) $x_1^h = \dfrac{\partial e_i(p,u_0)}{\partial p_1}$

$$\frac{\partial e_i(p,u_0)}{\partial p_i} = -\cancel{\frac{1}{r}}\, u_0 \left(p_1^{-r}+p_2^{-r}\right)^{-\frac{1}{r}-1} \cancel{(-r)}\; p_1^{-r-1}$$

$$= u_0 \left(\frac{1}{p_1^r}+\frac{1}{p_2^r}\right)^{-\frac{1}{\rho}} \quad p_1^{\frac{1}{\rho}-1}$$

$$= u_0 \left(\frac{p_1^r\, p_2^r}{p_1^r+p_2^r}\right)^{\frac{1}{\rho}} \quad p_1^{\frac{1}{\rho}-1}$$

$$= u_0 \left(\frac{p_2^r}{p_1^r+p_2^r}\right)^{1/\rho}$$

e) Misma solución que derivando.

Ejercicio 2

$$u(x) = x_1 + \sqrt{x_2}$$

La solución de esquina solo puede estar en x_1, entonces

$$\mathcal{L} = -P_1 x_1 - P_2 x_2 + \lambda(x_1 + \sqrt{x_2} - u_0) + \alpha x_1$$

C.P.O

$[x_1]: \quad -P_1 + \lambda + \alpha = 0 \qquad ①$

$[x_2]: \quad -P_2 + \dfrac{\lambda}{2} \dfrac{1}{\sqrt{x_2}} = 0 \qquad ②$

$[\lambda]: \quad x_1 + \sqrt{x_2} = u_0 \qquad ③$

Condición complementaria: $\alpha x_1 \geq 0$.

Caso 1: $\alpha > 0$, $x_1 = 0$.

③ $x_2 = u_0^2$

① en ② $\quad P_2 = \dfrac{P_1 - \alpha}{2} \dfrac{1}{u_0}$

$$\alpha = P_1 - 2P_2 u_0 > 0.$$

$$\Rightarrow \frac{P_1}{2P_2} > u_0$$

Caso 2 $\quad \alpha = 0, \quad x_1 > 0$

① $\Rightarrow \lambda = P_1$

② $\Rightarrow P_2 = \dfrac{P_1}{2} \dfrac{1}{\sqrt{x_2}} \Rightarrow x_2^h = \dfrac{1}{4}\left(\dfrac{P_1}{P_2}\right)^2$

③ $\quad x_1 = M_0 - \dfrac{1}{2} \dfrac{P_1}{P_2}$

Si $x_1 > 0 \Rightarrow \quad M_0 > \dfrac{1}{2} \dfrac{P_1}{P_2}$

Entonces

$$x_1^h(P, M_0) = \begin{cases} 0 & \text{si } M_0 < \frac{1}{2}\frac{P_1}{P_2} \\ M_0 - \frac{1}{2}\frac{P_1}{P_2} & \text{si caso contrario} \end{cases}$$

$$x_2^h(P, M_0) = \begin{cases} M_0^2 & \text{si } M_0 < \frac{1}{2}\frac{P_1}{P_2} \\ \frac{1}{4}\left(\frac{P_1}{P_2}\right)^2 & \text{caso contrario} \end{cases}$$

Pregunta 3

$$e(P, \mathcal{U}_0) = \left(4P_1^2 + 4P_2^2 + 3P_3^2 + 3P_4^2\right)^{1/2} \mathcal{U}_0$$

a) $x_1^h(P, \mathcal{U}_0) = \dfrac{\partial e(P, \mathcal{U}_0)}{\partial P_i}$

$$= \frac{\mathcal{U}_0}{2} \left(4P_1^2 + 4P_2^2 + 3P_3^2 + 3P_4^2\right)^{-1/2} 8P_1$$

$$= 4\,\mathcal{U}_0\, P_1 \left(4P_1^2 + 4P_2^2 + 3P_3^2 + 3P_4^2\right)^{-1/2}$$

b) $e(P, V(P,y)) = y$

$$\left(4P_1^2 + 4P_2^2 + 3P_3^2 + 3P_4^2\right)^{1/2} V(P,y) = y.$$

$$V(P,y) = y\left(4P_1^2 + 4P_2^2 + 3P_3^2 + 3P_4^2\right)^{-1/2}$$

c) $x_i^*(P,y) = x^h(P, V(P,y))$

$$= 4y\left(4P_1^2 + 4P_2^2 + 3P_3^2 + 3P_4^2\right)^{-1}$$

Ejercicio 4

a) ① $V(P_1+t, P_2, y+z^h(t)) = V^0$

$e(P_1+t, P_2, V^0) = y + z^h(t)$ $\Big\}$ cualquiera ① o ② están bien

② $z^h(t) = e(P_1+t, P_2, V^0) - y$.

b) $P_1 x_1^0 + P_2 x_2^0 = y$ $\{$ antes intervención

$(P_1+t)x_1^0 + P_2 x_2^0 = y + z^s(t)$ $\{$ Después intervención

$t x_1^0 = z^s(t)$.

c) Queremos mostrar $V(P_1+t, P_2, y+z^s(t)) \geq V(P_1+t, P_2, y+z^h(t))$

Notar que $V(P_1+t, P_2, y+z^s(t)) \geq u(x^0)$.

$\Rightarrow$ dado que al menos puede costear x^0.

$\Rightarrow V(P_1+t, P_2, y+z^s(t)) \geq V^0 = V(P_1+t, P_2, y+z^h(t))$

d) $\dfrac{\partial z^s(t)}{\partial t} = x_1^0$

$$\frac{\partial z^h(t)}{\partial t} = \frac{\partial e(P_1+t, P_2, V^0)}{\partial P_1+t} \cdot (1) = x_1^h(P_1+t, P_2, u_0)$$

$$\left.\frac{\partial z^h(t)}{\partial t}\right|_{t=0} = x_1^h(P_1, P_2, u_0) = x_1^0$$